

Cloud Computing (Practical 1)

1. AIM

To launch an EC2 instance on AWS and host a static website using Apache web server.

2. OBJECTIVE

To understand cloud computing services and deploy a simple static website on AWS EC2 instance.

3. THEORY

Cloud computing is a technology that allows users to access computing resources such as servers, storage, networking, and software over the internet instead of using local machines. It provides on-demand services with scalability, flexibility, and cost efficiency.

Cloud computing is mainly divided into three service models:

- **Infrastructure as a Service (IaaS)**
- **Platform as a Service (PaaS)**
- **Software as a Service (SaaS)**

In this experiment, **Amazon EC2 (Elastic Compute Cloud)** is used, which falls under the **IaaS model**.

Amazon Web Services (AWS)

Amazon Web Services (AWS) is a cloud computing platform provided by Amazon. It offers various services such as computing power, storage, databases, networking, and security.

AWS allows users to:

- Launch virtual servers
- Deploy applications
- Store and manage data
- Scale resources based on demand

Amazon EC2 (Elastic Compute Cloud)

Amazon EC2 is a web service that provides resizable virtual servers called **instances**. These instances allow users to run applications in the cloud.

Features of EC2:

- Scalable computing capacity
- Pay-as-you-go pricing
- Secure and reliable infrastructure
- Multiple operating systems support

Each EC2 instance acts like a virtual computer where users can install software and run applications.

EC2 Instance Components

An EC2 instance consists of the following components:

1. AMI (Amazon Machine Image)

An AMI is a template that contains the operating system and required software.
Example: Amazon Linux 2023

2. Instance Type

Defines the hardware configuration such as CPU, memory, and performance.
Example: t3.micro (free tier eligible)

3. Key Pair

Used for secure login into the instance. It contains:

- Public key (stored in AWS)
- Private key (downloaded by user)

4. Security Group

Acts as a virtual firewall controlling inbound and outbound traffic.

Example rules:

- SSH (Port 22) → Remote access
 - HTTP (Port 80) → Web access
-

5. Storage (EBS)

Elastic Block Store provides storage to EC2 instances.

Apache Web Server

Apache is an open-source web server used to host websites. It processes HTTP requests and serves web pages to users.

In this experiment:

- Apache is installed on EC2
- A static HTML page is created
- The website is accessed using public IP

Static Website Hosting

A static website consists of fixed content such as HTML, CSS, and images. It does not require server-side processing.

Steps involved:

- Create HTML file
- Place it in web server directory
- Access via browser using IP address

Working of the System

- EC2 instance is launched
- Apache server is installed
- Webpage is stored in server directory
- User accesses website using public IP
- Apache serves the webpage

Advantages of Using EC2

- Easy to launch and manage servers
- Cost-effective (pay per use)
- Scalable resources
- High availability and reliability

4. IMPLEMENTATION WITH OUTPUT

Step 1: Open AWS EC2 Dashboard

The screenshot displays the AWS Management Console for the EC2 service in the Europe (Stockholm) region. The interface includes a left-hand navigation menu with categories like EC2, Images, Elastic Block Store, and Network & Security. The main content area is divided into several panels:

- Resources:** A table showing the usage of various Amazon EC2 resources. The data is as follows:

Resource	Count
Instances (running)	0
Dedicated Hosts	0
Key pairs	1
Security groups	2
Auto Scaling Groups	0
Elastic IPs	0
Load balancers	0
Snapshots	0
Capacity Reservations	0
Instances	1
Placement groups	0
Volumes	1
- Launch instance:** A section with a "Launch instance" button and a "Migrate a server" link. It includes a note that instances will launch in the Europe (Stockholm) Region.
- Instance alarms:** A section showing the status of alarms. It indicates "0 in alarm", "0 OK", and "0 insufficient data". There is a "View in CloudWatch" link.
- Scheduled events:** A section at the bottom left for managing scheduled events.
- Service health:** A section showing the status of the EC2 service. It indicates that the service is operating normally in the Europe (Stockholm) region. It also lists available zones: eu-north-1a, eu-north-1b, and eu-north-1c.
- EC2 cost:** A section showing the cost of EC2 resources. It indicates that costs are covered by credits, with a remaining credit of \$100 USD and 180 days remaining.
- Account attributes:** A section showing account information, including the default VPC and settings.

Explanation:

The AWS EC2 dashboard is opened to manage cloud resources such as instances, volumes, and security groups.

Step 2: Configure EC2 Instance

Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices.

Name and tags

Name: practical 1

Application and OS Images (Amazon Machine Image)

Search our full catalog including 1000s of application and OS images

Amazon Linux 2023 kernel-6.1 AMI

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.11...

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Launch instance

Explanation:

A new EC2 instance is configured by selecting:

- Amazon Linux 2023 AMI
- t3.micro instance type
- Default storage

This defines the virtual server setup.

Step 3: Configure Security Group

Network settings

Network: vpc-056d9be565c2572e3

Subnet: No preference (Default subnet in any availability zone)

Auto-assign public IP: Enable

Firewall (security groups)

Create security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

- Allow SSH traffic from Anywhere
- Allow HTTPS traffic from the internet
- Allow HTTP traffic from the internet

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Summary

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Launch instance

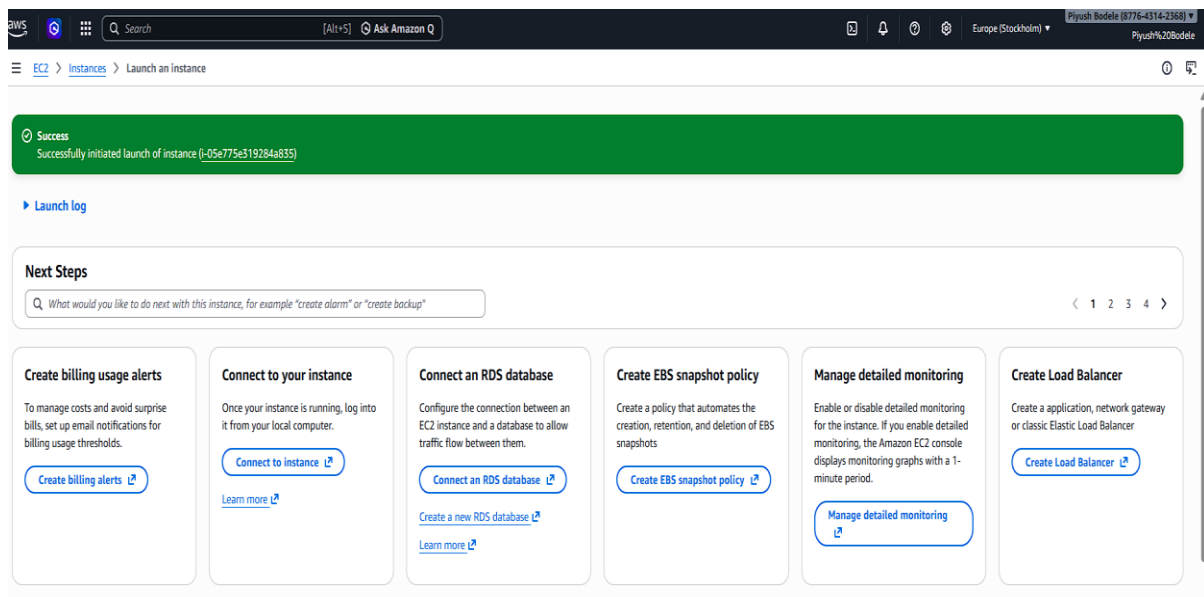
Explanation:

Security group rules are configured to allow:

- SSH (port 22) → for remote access
- HTTP (port 80) → to access website

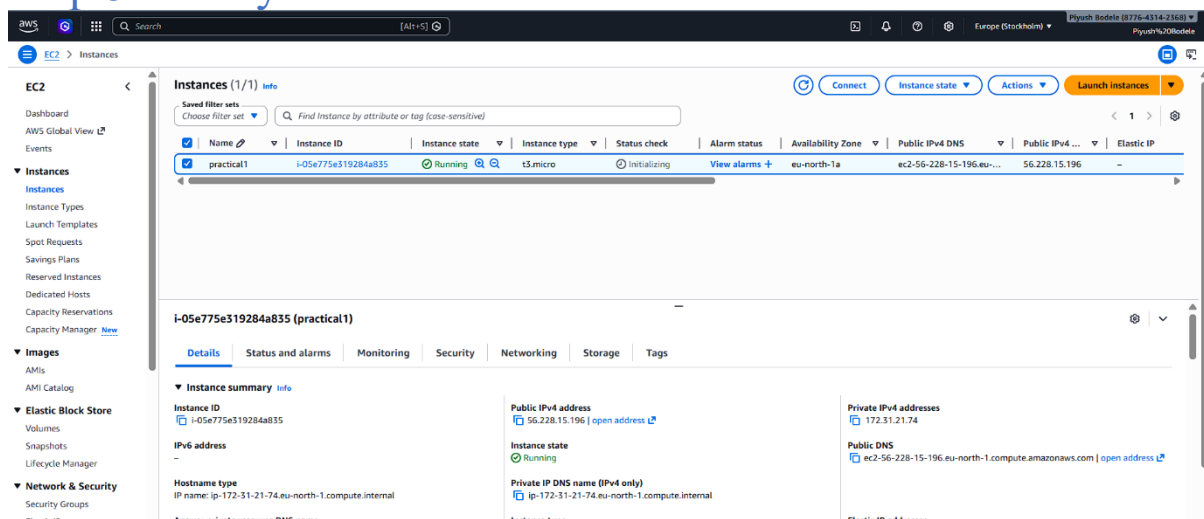
This ensures proper connectivity.

Step 4: Launch EC2 Instance

**Explanation:**

The instance is successfully launched, indicating that the virtual machine is created and ready.

Step 5: Verify Instance Status

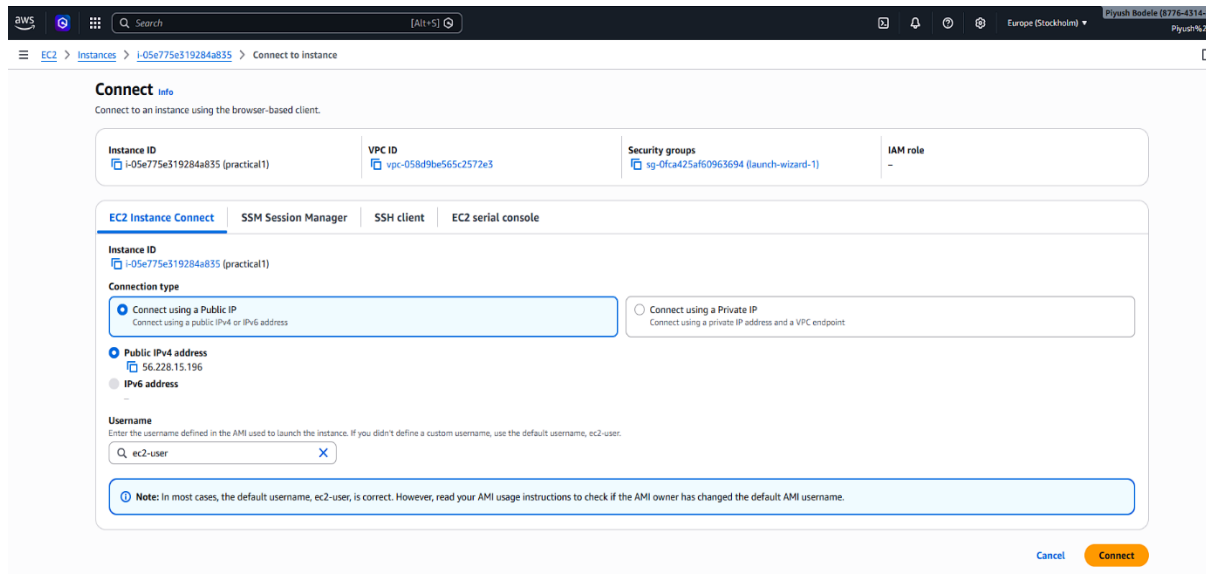


Explanation:

The instance status is checked and confirmed as **Running**.

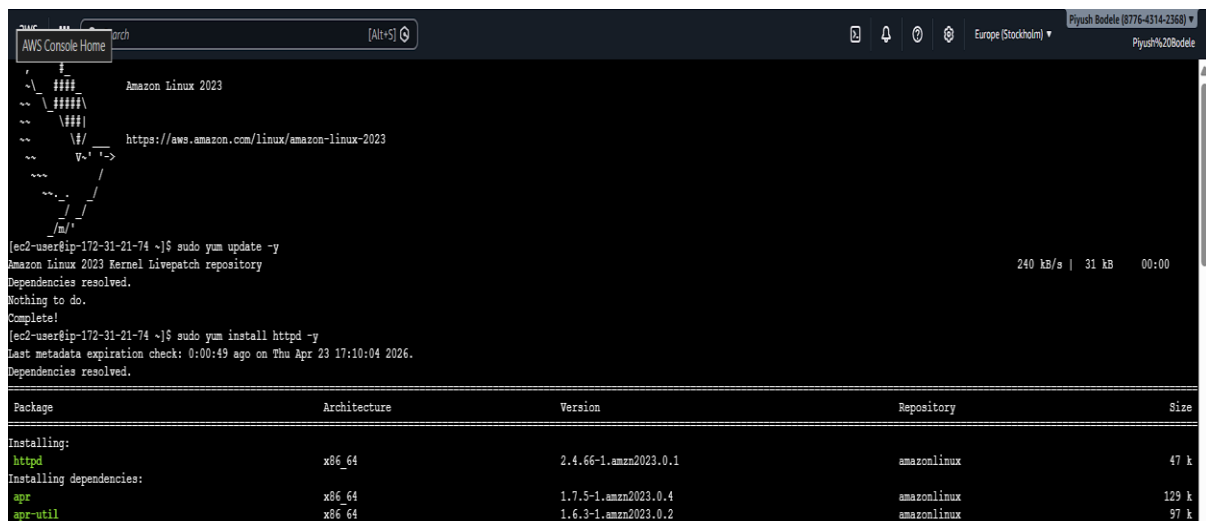
The public IP address is noted for accessing the website.

Step 6: Connect to EC2 Instance

**Explanation:**

The instance is connected using EC2 Instance Connect with username **ec2-user**, providing terminal access.

Step 7: Install Apache Server



```

aws [Search] [Alt+S] Europe (Stockholm) Piyush Bodele (8776-4314-2368) Piyush%20Bodele

Transaction Summary

Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm 2.8 MB/s | 97 kB 00:00
(2/13): apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64.rpm 356 kB/s | 13 kB 00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 2.9 MB/s | 129 kB 00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm 583 kB/s | 15 kB 00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 700 kB/s | 19 kB 00:00
(6/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm 1.7 MB/s | 47 kB 00:00
(7/13): httpd-core-2.4.66-1.amzn2023.0.1.x86_64.rpm 43 MB/s | 1.4 MB 00:00
(8/13): httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch.rpm 407 kB/s | 13 kB 00:00
(9/13): httpd-tools-2.4.66-1.amzn2023.0.1.x86_64.rpm 2.4 MB/s | 81 kB 00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm 11 MB/s | 315 kB 00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm 1.1 MB/s | 33 kB 00:00
(12/13): mod_http-2.0.27-1.amzn2023.0.3.x86_64.rpm 5.7 MB/s | 166 kB 00:00
(13/13): mod_lua-2.4.66-1.amzn2023.0.1.x86_64.rpm 2.5 MB/s | 60 kB 00:00

Total 12 MB/s | 2.4 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Installing : apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64 2/13
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
Installing : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
Installing : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 6/13
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13

i-05e775e319284a835 (practical1)
PublicPis: 56.228.15.196 PrivatePis: 172.31.21.74

```

Explanation:

Apache HTTP server is installed using the command:

- `sudo yum install httpd -y`

The installation process downloads and installs required packages successfully.

Step 8: Start and Enable Apache Server

```

AWS [Search] [Alt+S] Europe (Stockholm) Piyush Bodele (8776-4314-2368) Piyush%20Bodele
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.2.x86_64      apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64      apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.66-1.amzn2023.0.1.x86_64      httpd-core-2.4.66-1.amzn2023.0.1.x86_64      httpdfilesystem-2.4.66-1.amzn2023.0.1.noarch
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64      libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-21-74 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-21-74 ~]$ sudo systemctl enable httpd
sudo: systemctl: command not found
[ec2-user@ip-172-31-21-74 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-21-74 ~]$ systemctl status httpd
• httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-23 17:12:32 UTC; 1min 58s ago
     Docs: man:httpd.service(8)
  Main PID: 26024 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1014)
  Memory: 13.5M
    CPU: 173ms
  OSGroup: /system.slice/httpd.service
           └─26024 /usr/sbin/httpd -DFOREGROUND
             └─26025 /usr/sbin/httpd -DFOREGROUND
               └─26026 /usr/sbin/httpd -DFOREGROUND
                 └─26027 /usr/sbin/httpd -DFOREGROUND
                   └─26041 /usr/sbin/httpd -DFOREGROUND

Apr 23 17:12:32 ip-172-31-21-74.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Apr 23 17:12:32 ip-172-31-21-74.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Apr 23 17:12:32 ip-172-31-21-74.eu-north-1.compute.internal httpd[26024]: Server configured, listening on: port 80
[ec2-user@ip-172-31-21-74 ~]$ cd /var/www/html
[ec2-user@ip-172-31-21-74 html]$ sudo rm -rf *
[ec2-user@ip-172-31-21-74 html]$ sudo nano index.html
[ec2-user@ip-172-31-21-74 html]$

```

i-05e775e319284a835 (practical1)
PublicIPs: 56.228.15.196 PrivateIPs: 172.31.21.74

Explanation:

The server is started and enabled using:

- `sudo systemctl start httpd`
`sudo systemctl enable httpd`

The status shows **active (running)**, confirming successful execution.

Step 9: Create HTML Webpage



```
GNU nano 8.3 index.html Modified
<!DOCTYPE html>
<html>
<head>
  <title>My First Cloud Website</title>
</head>
<body style="text-align:center; font-family:Arial;">

  <h1>Welcome to My Website </h1>
  <h2>Hosted on AWS EC2</h2>

  <p>Name: Piyush</p>
  <p>Course: Computer Engineering</p>

  <p>This is my Cloud Computing Practical.</p>

</body>
</html>

File Name to Write: index.html
^G Help          ^O DOS Format    ^R Append       ^B Backup File
^C Cancel        ^M Mac Format    ^F Prepend      ^E Browee

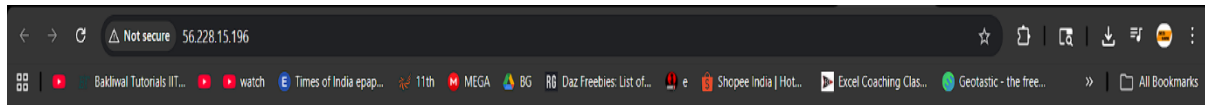
i-05e775e319284a835 (practical1)
PublicIPs: 56.228.15.196 PrivateIPs: 172.31.21.74
```

Explanation:

A static HTML file (index.html) is created in /var/www/html directory using nano editor.

This file contains the webpage content.

Step 10: Access Hosted Website



Welcome to My Website 🚀

Hosted on AWS EC2

Name: Piyush

Course: Computer Engineering

This is my Cloud Computing Practical.

Explanation:

The website is accessed using the public IP address in a browser.

The output confirms that the static website is successfully hosted.

5. CONCLUSION

The static website is successfully deployed and accessed using AWS EC2.